Kaiwen (Keven) Wu

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EDUCATION

University of California, San Diego Expected Jun 2029
B.S. in Mathematics-Computer Science

Coursework: Advanced Data Structures and Algorithms, Design and Analysis of Algorithms, Systems Programming, Computer Organization, Discrete Mathematics and Graph Theory, Linear Algebra

Honors and Competitions: AIME II: 10/15 (2024); AMC 12: Top 5% (2024); 1 year of competitive programming in C++.

EXPERIENCE

Triton Trading Group Oct 2026 – Present
Quantitative Developer *California, United States*

- Developing trading algorithms, backtesting systems, and tools for the division's quantitative research.
- Building low latency systems and encapsulating fast calculations of factors using modern C++.
- Continuously optimizing a trading strategy on US equities using ATR, EMA, momentum factor, and Kelly sizing.

Dubbing AI Jul 2026 – Aug 2026
Software Engineer Intern *Shanghai, China*

- Enabled long-term memory in a real-time voice agent by integrating authenticated extraction, scoped retrieval, and lifecycle APIs with a C++/Go stack.
- Achieved **100% knowledge Recall@1**, **99.5% slang Recall@1**, and **13.36 ms p95 retrieval latency** in retrieval evaluations by optimizing indexes with hash maps and hierarchical lookup.
- Implemented a Redis-backed short-term low-latency memory cache for the company's agent product, supporting memory access during RAG retrieval to reduce response latency.

AscendEX Aug 2024 – Sep 2024
Data Analysis Intern *New York, United States*

- Identified transaction patterns and anomalies across **300,000+ records** using Python and pandas; refined K-means clustering and implemented sliding-window thresholds for USDT transactions.
- Presented analysis methods and findings to the CEO and board members in a technical report.

PROJECTS

SWE Agent: Coding Agent Co-designed with Its Serving Layer Oct 2026 – Present | GitHub

- Building a terminal-first coding agent in Python with no agent frameworks, implementing tool use over raw HTTP (httpx); a tagged release per phase with pyright, ruff, pytest, gitleaks, and GitHub Actions CI.
- Implementing the agent loop (model call $\rightarrow$ tool execution $\rightarrow$ observation) with step, cost, and wall-time budgets; each step is logged as a JSONL trace (tokens, cache counters, tool timings) for KV-cache analysis.
- Next: SWE-bench evals against a mini-swe-agent baseline, then a deterministic KV-cache simulator, an offline-optimal min-cost-flow bound on agent traces, and agent-aware KV retention in a nano-vllm fork.

Adaptive Grid Trading Strategy GitHub

- Built an adaptive Binance spot strategy in Python that screened high-volume USDT pairs and ranked candidates using 30-minute log-return volatility and AR(1) mean-reversion signals.
- Self-studied stochastic calculus (GBM, OU processes, Ito's lemma); fit OU parameters and optimized grid spacing with SciPy; built Backtrader simulations and stateful ccxt live execution with z-score sizing, modeled costs, and risk controls.

Systems Programming Projects UCSD coursework

- Built a Unix-like shell in C with interactive and batch modes, built-in commands, and process control via fork(), execvp(), and wait(), with system-call error handling.
- Implemented an ARM fetch-decode-execute simulator and a linked-list C allocator that passed Valgrind.

TECHNICAL SKILLS

Languages: C++, C, Java, Python, Go, Rust, SQL

Backend and Tools: FastAPI, REST APIs, Redis, SQLite, Docker, Git, Linux, pytest, GitHub Actions (CI/CD), LaTeX

AI and Data: LLM tool use, agent loops, RAG, embeddings, reranking, pandas, NumPy, SciPy, statsmodels, scikit-learn, ccxt

Security Foundations: TCP/IP, HTTP, GDB, objdump, radare2, tcpdump, nmap, netdiscover, ettercap, hping